

MacroOps AI: A GenAI-Powered Fulfillment and Last-Mile Operations Intelligence Platform

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Abstract—Modern quick-commerce and e-commerce operations involve tightly coupled activities including order processing, inventory management, picking, workforce coordination, rider assignment, and last-mile delivery. Operational teams often need to identify exceptions quickly, understand their causes, assess Service Level Agreement (SLA) risk, and decide corrective actions. This paper presents MacroOps AI, a prototype GenAI-powered fulfillment and last-mile operations intelligence platform designed to support these activities through a unified operational dashboard. The implemented prototype combines structured operational analytics, KPI monitoring, exception detection, rule-based SLA risk classification, a project-level Retrieval-Augmented Generation (RAG) knowledge assistant, a GenAI Operations Copilot, and a frontline AI assistant. The implementation uses Python, Pandas, Streamlit, Plotly, OpenAI GenAI, CSV-based operational datasets, and a lightweight keyword-based retrieval approach for the available SOP knowledge base. The prototype processes 100,000 orders and 40,000 inventory records and reports operational indicators including SLA breach rate, average delivery delay, inventory accuracy, picking time, workforce backlog, and detected exceptions. The work demonstrates an integrated MVP approach for moving operations management from manual and reactive investigation toward data-driven and AI-assisted decision support while retaining human approval for operational actions.

Index Terms—Generative AI, operations intelligence, quick-commerce, last-mile delivery, SLA monitoring, exception management, RAG, operations copilot, Streamlit, business analytics

I. INTRODUCTION

Quick-commerce, e-commerce, and retail fulfillment networks operate under strict time commitments and changing operational conditions. Orders must be processed, inventory must remain accurate, items must be picked efficiently, riders must be allocated, and deliveries must be completed within promised time windows. Delays in one part of the process can propagate to downstream activities and create SLA breaches, customer dissatisfaction, additional workload, and operational cost.

The MacroOps AI project addresses the need for an integrated operations intelligence layer that can answer five practical questions: what is happening, why it is happening, what is likely to happen next, what action should be considered, and how AI can assist both managers and frontline teams. The project roadmap supplied for Phase 03 describes a progression from business discovery and product strategy to data and

analytics, AI/GenAI, MVP development, integration, testing, UAT, evaluation, deployment, and business-impact assessment.

This paper documents the implemented prototype, its architecture, data foundation, analytics, exception workflows, SLA risk classification, RAG knowledge assistance, GenAI capabilities, dashboard design, and observed prototype results. The paper distinguishes implemented prototype behavior from planned or future capabilities.

II. PROBLEM STATEMENT

Dynamic fulfillment and last-mile networks can experience delivery delays, inventory discrepancies, stockouts, workforce constraints, rider-assignment delays, picking bottlenecks, and SLA risks. When operational information is distributed across datasets, dashboards, spreadsheets, and procedural documents, investigation can become manual and reactive.

The central problem addressed by MacroOps AI is therefore:

How can operational data, analytics, GenAI, and workflow-oriented intelligence be combined into a unified prototype that helps operations teams monitor performance, detect exceptions, understand risks, access SOP guidance, and make faster decisions?

III. OBJECTIVES

The project objectives are:

- 1) Monitor orders, inventory, picking, workforce, and delivery performance through consolidated KPIs.
- 2) Detect operational exceptions including SLA breaches, stockouts, delivery assignment delays, picking delays, and workforce backlog.
- 3) Analyze operational patterns and provide contextual explanations.
- 4) Classify SLA risk using operational delivery-delay information.
- 5) Provide natural-language assistance through a GenAI Operations Copilot.
- 6) Ground operational guidance using a project-level SOP knowledge base.
- 7) Support frontline personnel with contextual task and exception assistance.
- 8) Establish a framework for evaluating operational and business impact.

IV. MOTIVATION AND NEED

Fast fulfillment and reliable delivery commitments are important in quick-commerce and e-commerce. Managers require rapid visibility into bottlenecks and exceptions, while frontline teams need actionable guidance rather than lengthy manual searches through SOP documents.

The project therefore combines conventional analytics with GenAI assistance. The intended operating model is to move from manual data analysis and delayed issue identification toward AI-assisted monitoring, early risk identification, automated analysis, GenAI explanation, recommended actions, human approval, and recorded resolution. The human-in-the-loop principle is retained because operational recommendations should be reviewed by responsible personnel before action.

V. PROJECT ROADMAP AND SCOPE

The supplied Phase 03 project plan defines a three-month progression from September to November 2026.

TABLE I
THREE-MONTH PROJECT ROADMAP

Month	Major activities
September	Business discovery, product strategy, requirements, stakeholder analysis, PRD, MVP scope, planning.
October	Data and analytics, AI/GenAI, RAG, dashboard, SLA prediction, exception workflow, MVP development.
November	Integration, testing, UAT, GenAI evaluation, deployment, business impact, documentation and presentation.

The core MVP scope includes an Operations Dashboard, Order SLA Monitoring, Exception Management, RAG Assistant, and AI Operations Copilot. Enhanced capabilities include SLA prediction and frontline AI assistance; multilingual interaction, voice interaction, computer vision, and advanced multi-agent architecture remain future capabilities.

VI. LITERATURE AND TECHNOLOGY BACKGROUND

Operations intelligence systems commonly combine operational databases, business intelligence, predictive analytics, and workflow management. For this project, the supplied project specification emphasizes business analytics for KPIs and anomalies, machine-learning exploration for SLA risk, and Generative AI with RAG for natural-language interaction and grounded SOP assistance.

RAG is relevant to operational assistants because generation can be supported by retrieved organizational knowledge instead of relying only on model parameters. In the implemented prototype, however, the retrieval mechanism is intentionally lightweight: it uses keyword overlap to select the most relevant available text document. This is a prototype implementation rather than a vector-embedding retrieval system.

The project also uses Generative AI for natural-language explanations and recommendations. The intended design keeps AI assistance connected to operational context and approved knowledge while preserving human oversight.

VII. SYSTEM ARCHITECTURE

The MacroOps AI architecture is organized into four logical layers:

- 1) **Data Layer:** operational CSV datasets are loaded, validated, cleaned, and transformed into usable analytical tables.
- 2) **Analytics Layer:** KPIs, business metrics, exception rules, and operational insights are generated.
- 3) **AI Layer:** SLA risk classification, RAG retrieval, GenAI assistance, frontline assistance, and recommendations are provided.
- 4) **Application Layer:** Streamlit integrates the analytical and AI capabilities into a unified dashboard.

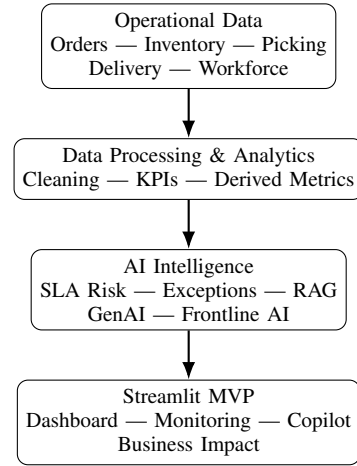


Fig. 1. MacroOps AI logical system architecture.

VIII. DATASET AND DATA PREPARATION

The prototype uses structured operational datasets representing orders, inventory, delivery, picking, and workforce activities. The order dataset contains fields including order identifier, store identifier, order time, item count, order value, promised time, actual delivery time, status, SLA-breach indicator, and delivery delay in minutes.

Data processing includes loading CSV files, converting applicable datetime fields, removing duplicates, calculating derived measures, and generating exception records. The resulting operational foundation supports dashboard KPIs and rule-based exception detection.

IX. BUSINESS ANALYTICS AND KPI FRAMEWORK

The analytics module calculates operational indicators such as total orders, order value, average order value, SLA breaches, SLA breach rate, average delivery delay, stockout rate, inventory accuracy, average distance, assignment delay, average pick duration, items picked or missing, completed tasks, pending tasks, and detected exceptions.

The dashboard is designed around the operational questions of performance monitoring and exception investigation rather than only visual presentation. This supports manager-oriented

TABLE II
OPERATIONAL DATA FOUNDATION

Dataset	Main analytical information
Orders	Order volume, value, promised/actual times, SLA status, delivery delay.
Inventory	Inventory records, stockouts, inventory accuracy.
Delivery	Delivery distance, rider assignment delay, delivery performance.
Picking	Pick duration, items picked and missing, productivity.
Workforce	Employees, completed tasks, pending tasks, workload backlog.

decision making by bringing related metrics into a common interface.

X. SLA MONITORING AND RISK CLASSIFICATION

SLA monitoring uses the delivery-delay field in the order dataset. In the current prototype, SLA risk is implemented as a transparent rule-based classification rather than a trained machine-learning model.

The implemented thresholds are:

$$Risk = \begin{cases} Low, & delay < 5 \text{ min} \\ Medium, & 5 \leq delay < 10 \text{ min} \\ High, & delay \geq 10 \text{ min} \end{cases} \quad (1)$$

The prototype produced 43,824 Low-risk, 33,753 Medium-risk, and 22,423 High-risk orders. These categories provide an operational prioritization mechanism. A future version can replace the rule-based classifier with a trained model using features such as order age, item count, picking duration, workload, inventory availability, rider availability, distance, and time of day, as envisaged in the project specification.

XI. EXCEPTION MANAGEMENT

The exception module consolidates operational conditions into an `exceptions.csv` output. Implemented rules include:

- SLA breaches based on the order SLA indicator.
- Inventory stockouts based on inventory status.
- Delivery assignment delay above 15 minutes.
- Picking delays above the 90th percentile.
- Workforce backlog above the 90th percentile.

The generated exception dataset contained 3,408 operational exceptions in the prototype run. Exception management is intended to reduce the need for manual inspection by bringing potentially important cases into a common operational view.

XII. RAG-BASED KNOWLEDGE ASSISTANCE

A project-level delivery-delay SOP was created as the available knowledge source. The SOP covers issue identification, delivery assignment checks, delivery conditions, prioritization, corrective action, resolution recording, and preventive action.

The current retrieval implementation reads text documents from the knowledge-base folder and compares words in the

user question with words in each available document. The document with the strongest keyword overlap is selected. The retrieved SOP content is then supplied as context to the GenAI assistant.

This is a deliberately simple RAG prototype. It should not be interpreted as a vector database or embedding-based semantic retrieval implementation. Future work can introduce document chunking, embeddings, a vector database, metadata filtering, source citations, and retrieval evaluation.

XIII. GENAI OPERATIONS COPILOT

The AI Operations Copilot provides a natural-language interface for operational questions. It can use operational context, KPI information, exception information, SLA risk interpretation, and retrieved SOP context to generate explanations and recommended actions.

The intended workflow is:

User Question → Operational Context → SOP Retrieval → GenAI → Exp

The system is designed so that recommendations remain decision support rather than autonomous operational commands. The responsible manager should review and approve operational actions.

XIV. FRONTLINE AI ASSISTANT

The frontline AI assistant is designed to support operational personnel with contextual task guidance, SOP assistance, and exception-resolution support. It complements the manager-facing dashboard by focusing on practical questions encountered during execution.

The separation between manager and frontline experiences follows the project requirement that manager interfaces emphasize KPIs, alerts, root-cause analysis, recommendations, and actions, while frontline interfaces emphasize tasks, AI assistance, SOP guidance, exceptions, and resolution.

XV. MVP DASHBOARD AND USER EXPERIENCE

The implemented Streamlit application contains the following pages:

- 1) Executive Dashboard
- 2) SLA Monitoring
- 3) Exception Management
- 4) Inventory Intelligence
- 5) Delivery Analytics
- 6) Picking Analytics
- 7) Workforce Analytics
- 8) Business Impact
- 9) Frontline AI Assistant
- 10) AI Operations Copilot

Plotly visualizations and tabular views are used where appropriate. The dashboard brings data analytics and AI assistance together so that users can move from KPI monitoring to exception investigation and then to contextual assistance.

XVI. AGILE AND SCRUM EXECUTION

The project follows the Agile/Scrum direction specified in the Phase 03 requirements. The development flow can be represented as:

Backlog → Sprint Planning → Development → Integration → Review → Feedback → Improvement.

The core MVP priority is P0: dashboard, SLA monitoring, exception management, RAG assistant, and AI Operations Copilot. P1 capabilities include SLA prediction, frontline AI, and multilingual interaction. P2 capabilities include voice interaction, computer vision, and advanced multi-agent architecture.

XVII. TESTING AND GENAI EVALUATION FRAMEWORK

The Phase 03 plan specifies functional, integration, API, data, GenAI, usability, and product testing, followed by UAT. The UAT sequence is prepare, execute, record, identify gaps, fix, retest, and obtain business approval/sign-off.

The project specification defines the following GenAI evaluation targets:

- Answer accuracy greater than 90%.
- Groundedness greater than 95%.
- Hallucination rate below 5%.
- Retrieval relevance greater than 90%.
- Response time below 5 seconds.
- User satisfaction greater than 4/5.

These are evaluation targets from the project requirements, not measured achievements claimed by the current prototype. Formal evaluation should be completed using a defined question set, reference answers, retrieved-source checks, response-time measurements, and user feedback.

XVIII. PROTOTYPE RESULTS

The current implementation provides the following observed operational indicators.

TABLE III
OBSERVED PROTOTYPE METRICS

Metric	Observed value
Total orders	100,000
SLA breach rate	89.67%
Average delivery delay	6.43 min
Inventory records	40,000
Inventory accuracy	89.42%
Stockout rate	0.01%
Average pick time	13.03 min
Pending workforce tasks	59,828
Operational exceptions	3,408

The results indicate that the prototype can process a substantial synthetic/operational dataset and surface multiple categories of operational information through a single application. The high SLA breach rate is retained as an observed dataset result rather than being modified to make the system appear better.

XIX. BUSINESS IMPACT FRAMEWORK

The project defines business-impact measures including exception resolution time, SLA breach rate, inventory accuracy, picker productivity, manager investigation time, employee training time, AI adoption, and operational decision-making speed.

The intended transformation is:

Manual Analysis → Delayed Identification → Manual Investigation → Review → Feedback → Improvement.

AI Monitoring → Early Risk Detection → Automated Analysis → GenAI

The current prototype establishes the measurement framework but does not claim a measured before-versus-after business improvement. Such an assessment requires controlled comparison, UAT data, or production observations.

XX. DEPLOYMENT AND PROJECT DELIVERABLES

The specified deployment flow is Code → GitHub → Build → Deploy → Test → Monitor → Go Live. Suitable platforms mentioned in the project requirements include Streamlit-based deployment and cloud application platforms.

The project deliverables include the working MVP, source code, datasets/sample data, exception dataset, SOP knowledge base, analytics modules, research paper on Overleaf, GitHub repository, project-management records, progress presentations, testing/UAT artifacts, deployment documentation, and business-impact assessment.

XXI. LIMITATIONS

The current prototype has several limitations:

- 1) SLA risk classification is rule-based and is not a trained predictive machine-learning model.
- 2) RAG retrieval uses keyword overlap rather than embeddings and vector search.
- 3) The knowledge base currently represents a project-level delivery-delay SOP rather than a broad enterprise document collection.
- 4) Formal UAT and quantitative GenAI evaluation are not claimed as completed results in this paper.
- 5) Business-impact improvements are defined as evaluation metrics but are not claimed as measured production outcomes.
- 6) The prototype is intended as an MVP and requires additional engineering for production-scale authentication, observability, security, deployment, and governance.

XXII. FUTURE SCOPE

Future development can extend the prototype in several directions:

- Train and validate an SLA-risk machine-learning model.
- Introduce embedding-based RAG with a vector database and source-level citations.
- Expand the knowledge base with approved SOPs and operational policies.

- Add multilingual and voice interaction.
- Explore computer vision for operational monitoring.
- Introduce advanced multi-agent workflows with controlled tool access.
- Add production monitoring, authentication, audit logging, and model governance.
- Conduct formal UAT and quantitative GenAI evaluation.
- Measure before-and-after business impact using controlled operational data.

XXIII. CONCLUSION

MacroOps AI demonstrates an integrated prototype for fulfillment and last-mile operations intelligence. The implementation combines operational analytics, KPI monitoring, exception management, SLA risk classification, RAG-based SOP assistance, GenAI Operations Copilot functionality, frontline AI support, and a Streamlit dashboard.

The prototype processes 100,000 orders and 40,000 inventory records and surfaces operational indicators including SLA breach rate, delivery delay, inventory accuracy, picking performance, workforce backlog, and exceptions. The work provides a practical foundation for moving from reactive operational investigation toward proactive, AI-assisted decision support. Further work is required to replace prototype-level rules and keyword retrieval with validated predictive models and semantic retrieval, complete formal UAT and GenAI evaluation, and establish production deployment and measured business impact.

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